

VALID: A 12-Item Validation Checklist for Financial Machine Learning

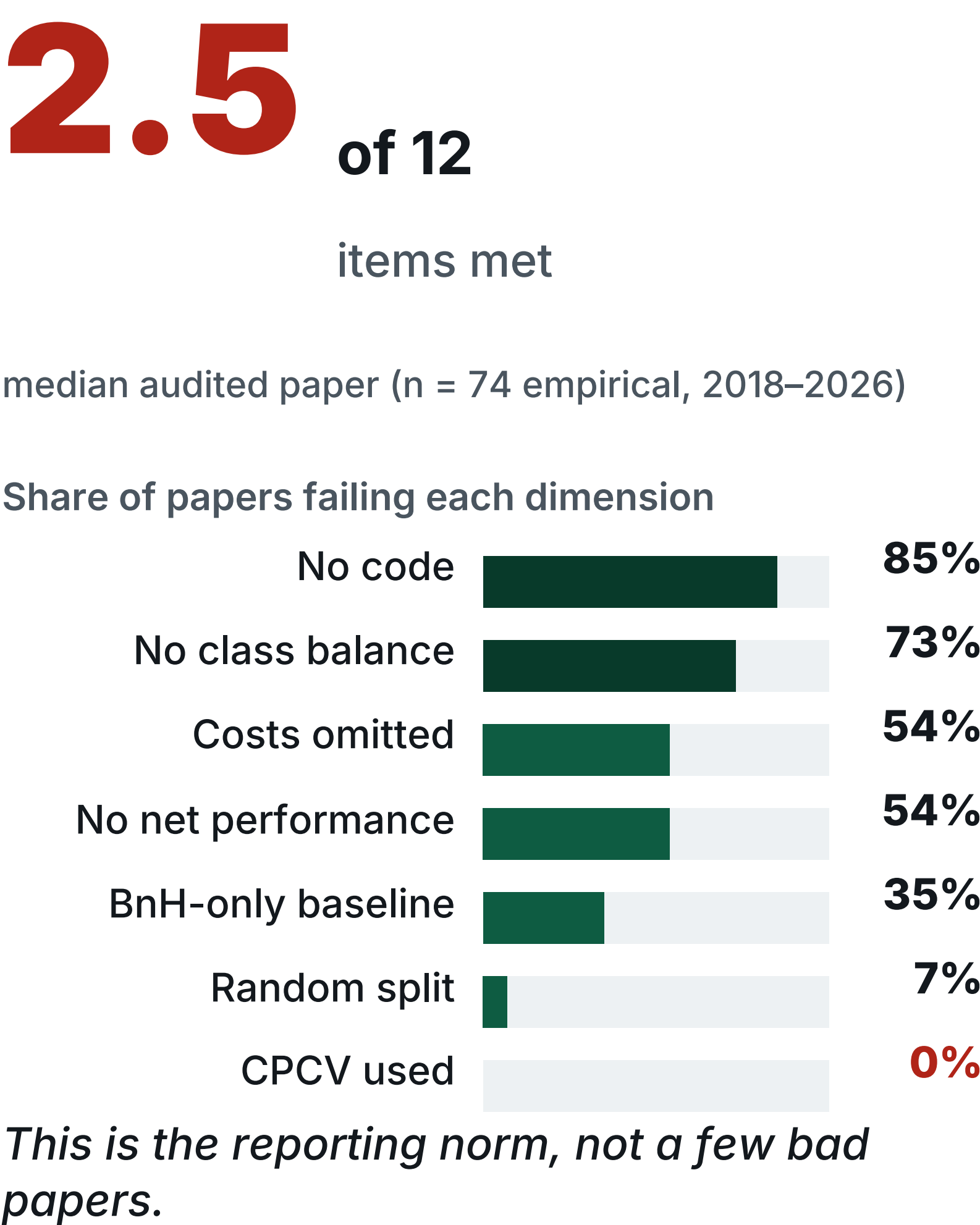
Jaewook Kim · Independent Researcher, JW Quant Research · KDD-MLF 2026, Jeju · Paper 18 (Oral)



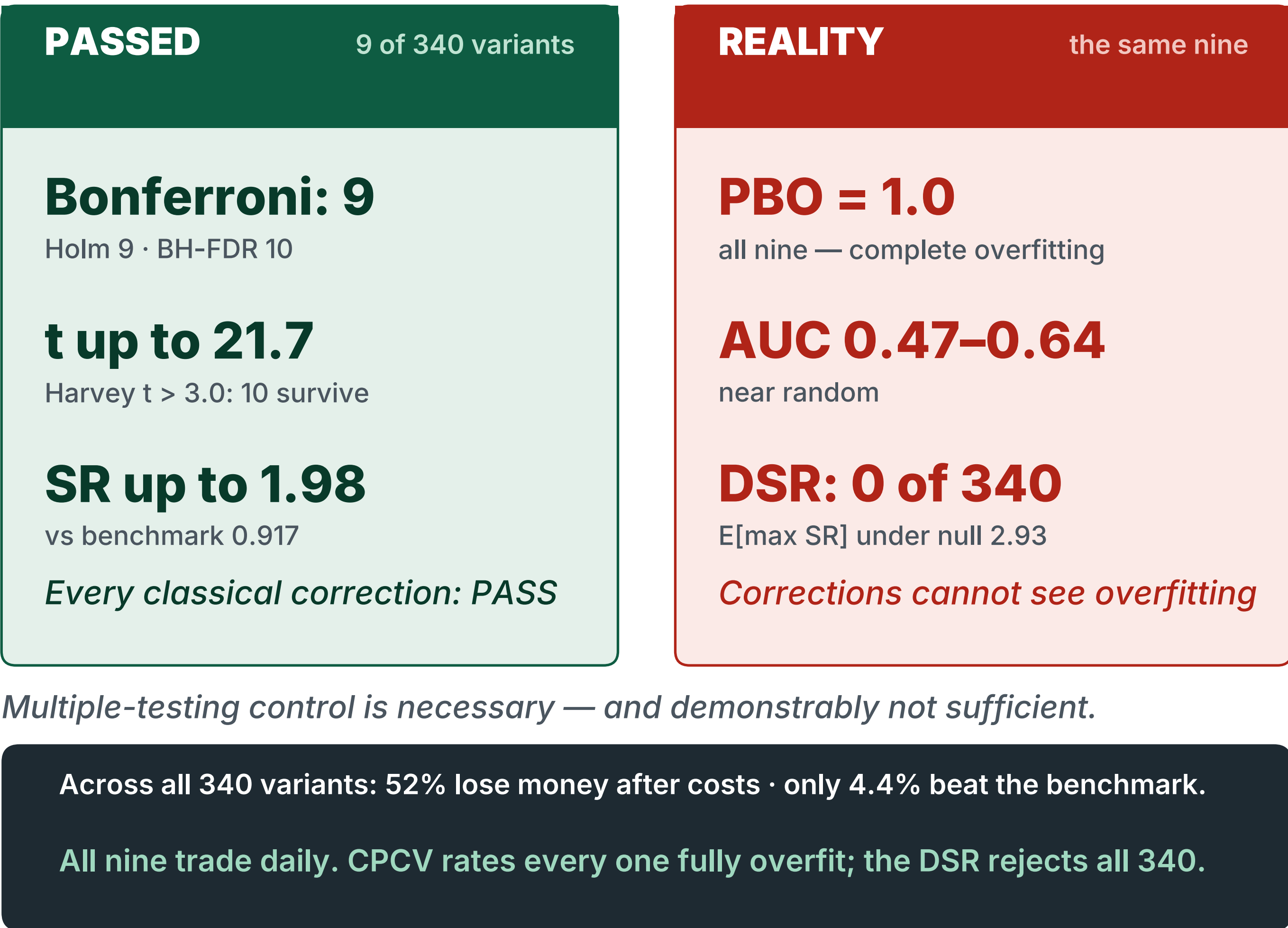
Statistically perfect.
Economically worthless.

340 strategy variants · 80 papers audited · one 12-item gate that tells them apart in ~5 minutes

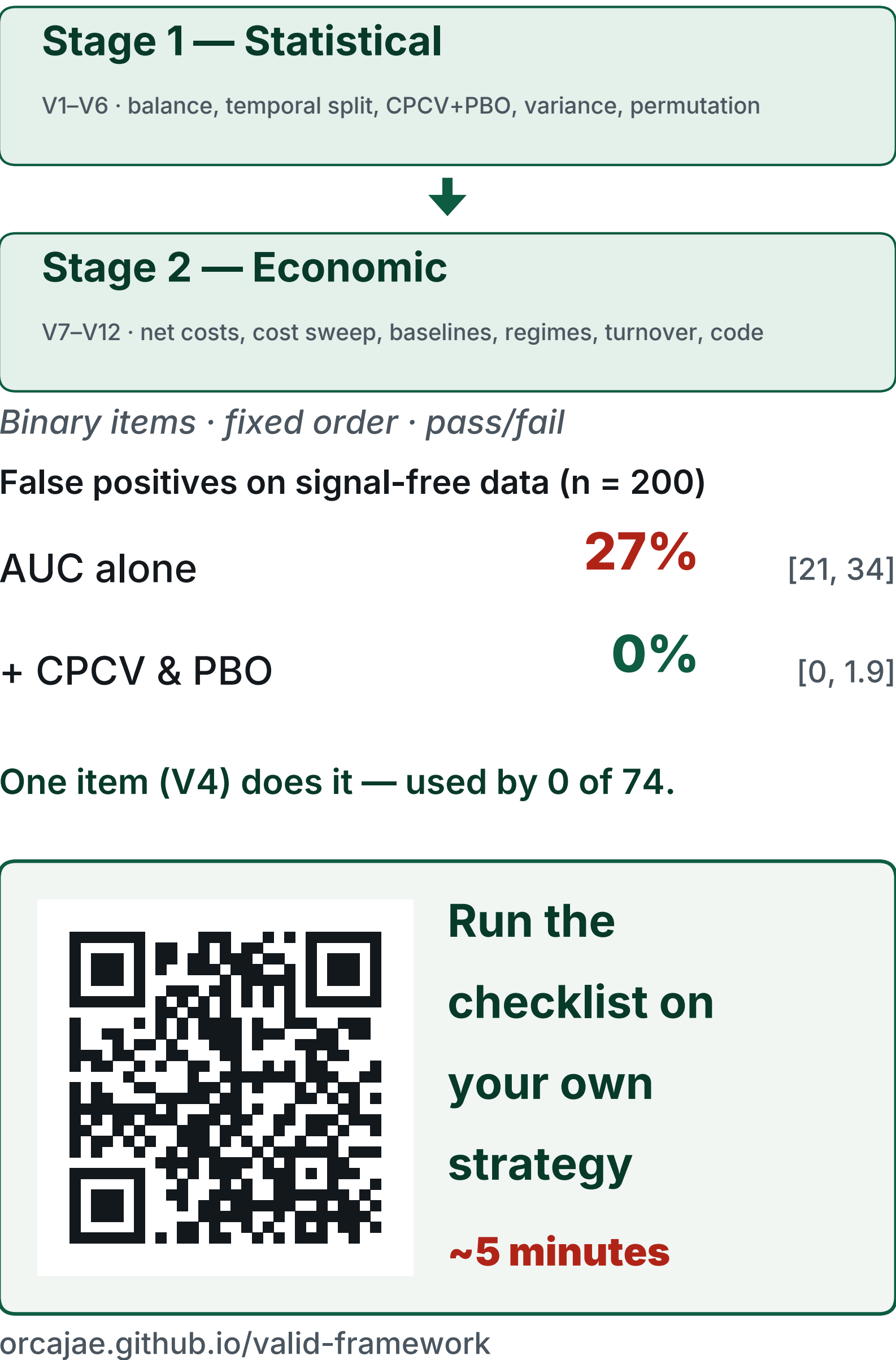
The Field Today



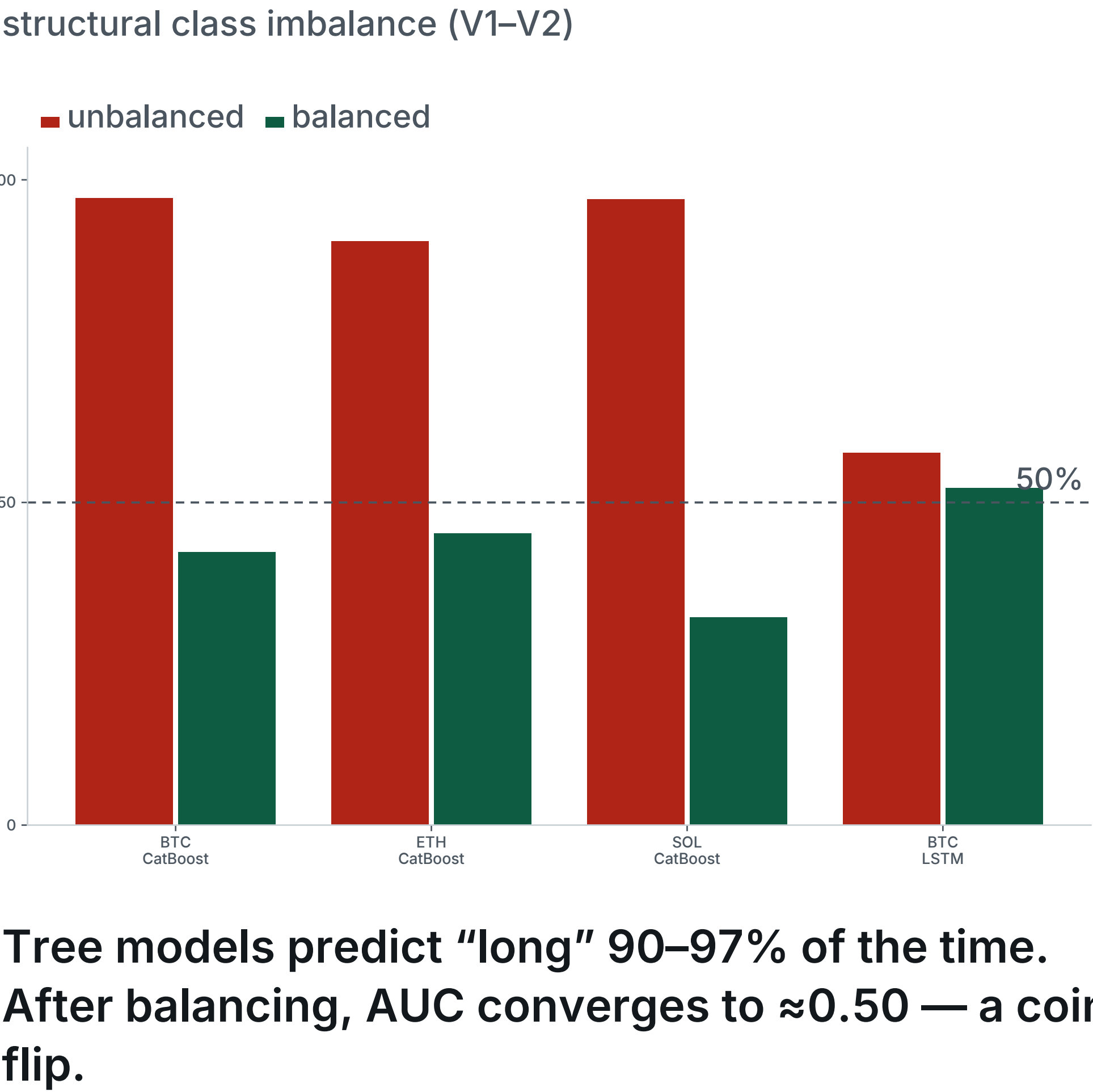
The Strategies That Passed Everything



The Gate



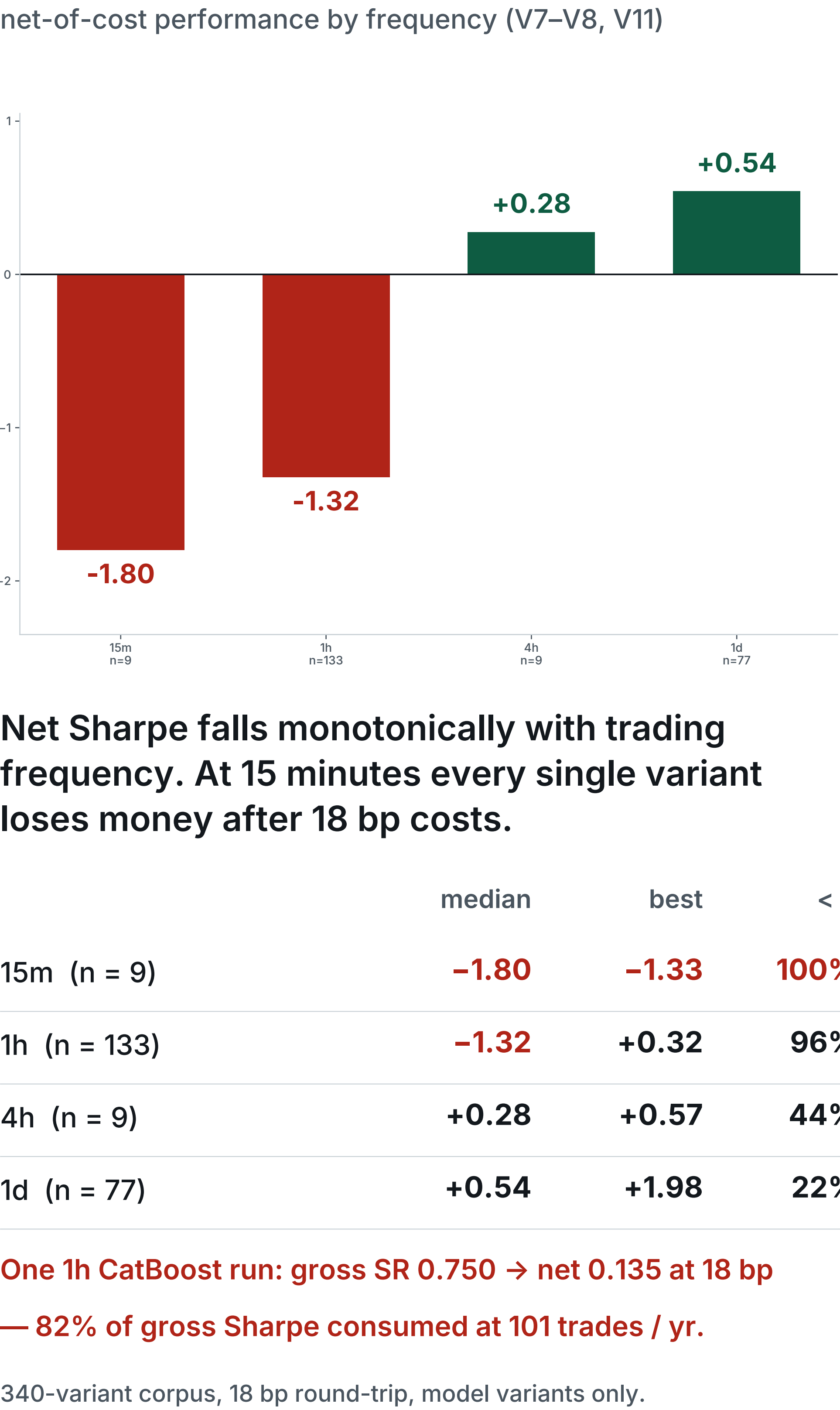
Bull Bias



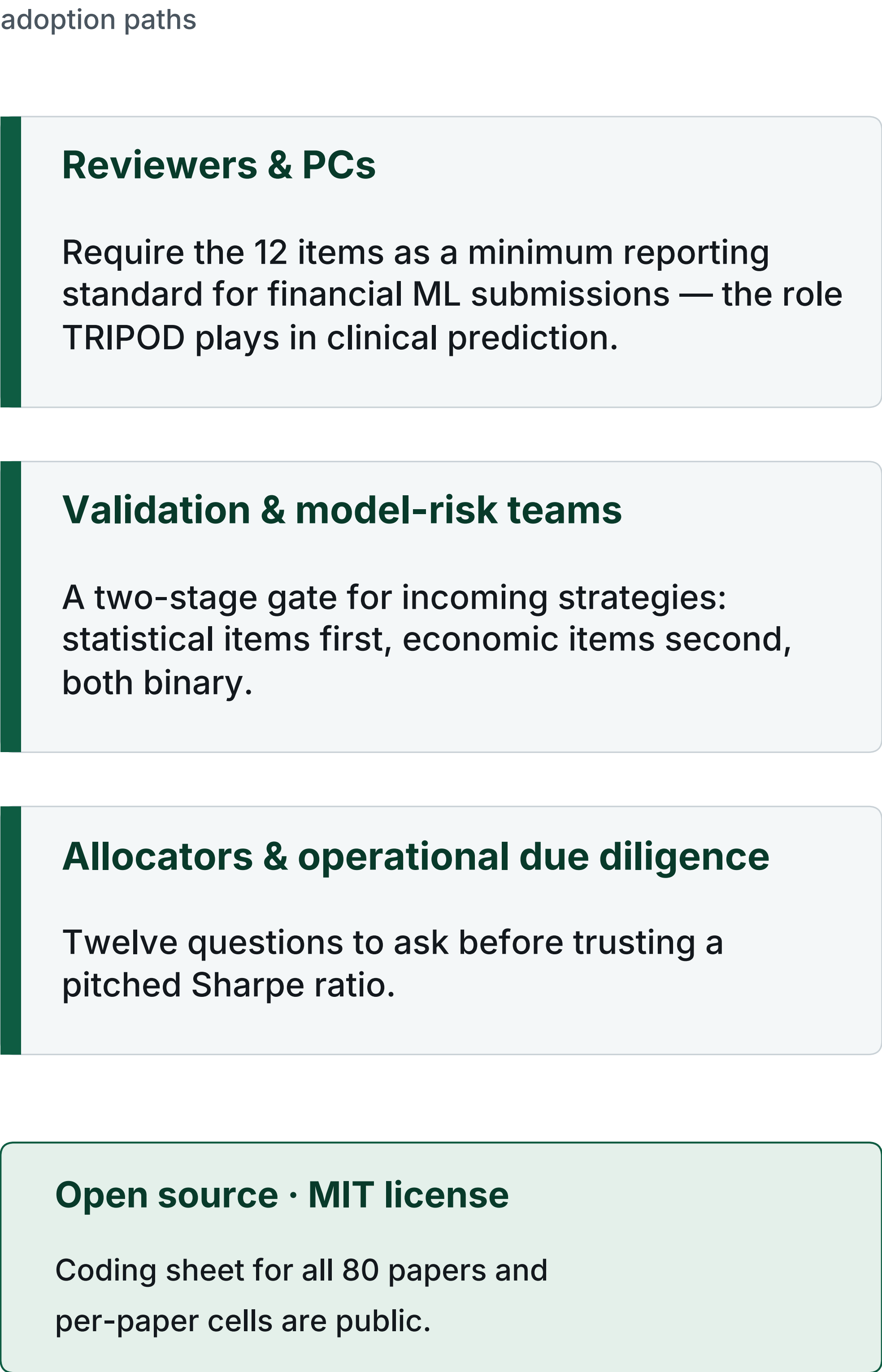
Balanced AUC (BTC / ETH / SOL)	0.50 · 0.49 · 0.51
Net SR, 1h balanced	−0.88 · −1.28 · −0.21
18 combinations, BTC mean long	85%
BTC mean balanced AUC / net SR	0.484 / −0.174

All 1h models: PBO = 1.000.

Cost Illusion



Who Is This For?



Score the last paper you reviewed

median in our audit: 2.5 / 12

- ☐ **V1** Report prediction class distribution
- ☐ **V2** Test with / without class balancing
- ☐ **V3** Use temporal splitting only
- ☐ **V4** Apply CPCV with PBO
- ☐ **V5** Report Var(SR_IS)
- ☐ **V6** Include permutation tests (≥100)
- ☐ **V7** Report net performance with costs
- ☐ **V8** Cost sensitivity analysis
- ☐ **V9** Compare against simple baselines
- ☐ **V10** Evaluate in bear markets and regime transitions
- ☐ **V11** Report trade frequency
- ☐ **V12** Provide code for reproducibility

V1–V6 Stage 1 (statistical) V7–V12 Stage 2 (economic)

v1.0 — not a certification. The checklist has had no formal Delphi review; community revision is invited.

Selected references

Harvey, Liu & Zhu (2016). ...and the Cross-Section of Expected Returns. RFS 29(1).
McLean & Pontiff (2016). Does Academic Research Destroy Return Predictability? JF 71(1).
Hou, Xue & Zhang (2020). Replicating Anomalies. RFS 33(5).
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Witzany (2021). A Bayesian Approach to Measurement of Backtest Overfitting. Risks 9(1).
Kapoor & Narayanan (2023). Leakage and the Reproducibility Crisis in ML-based Science. Patterns 4(9).